Solutions for GATE 2026 Electrical Engineering (EE)

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Solution for Q65

Q.65	The integral $\frac{1}{\pi}\int_0^{\infty}\frac{x^{2026}}{(1+x^{2026})(1+x^2)}\,dx$ evaluates to ________ (*Round off to two decimal places*)

Solution:
Evaluate

$$I=\frac{1}{\pi}\int_0^{\infty}\frac{x^{2026}}{(1+x^{2026})(1+x^2)}\,dx.$$

Using the substitution

$$x=\frac{1}{t},$$

we have

$$dx=-\frac{dt}{t^2}.$$

Therefore,

$$I=\frac{1}{\pi}\int_{\infty}^{0}\frac{\frac{1}{t^{2026}}}{\left(1+\frac{1}{t^{2026}}\right)\left(1+\frac{1}{t^2}\right)}\left(-\frac{dt}{t^2}\right)$$
$$=\frac{1}{\pi}\int_0^{\infty}\frac{1}{(1+t^{2026})(1+t^2)}\,dt.$$

Renaming the variable t to x,

$$I=\frac{1}{\pi}\int_0^{\infty}\frac{1}{(1+x^{2026})(1+x^2)}\,dx.$$

Now add this expression to the original one:

$$2I=\frac{1}{\pi}\int_0^{\infty}\left[\frac{x^{2026}}{(1+x^{2026})(1+x^2)}+\frac{1}{(1+x^{2026})(1+x^2)}\right]dx$$
$$=\frac{1}{\pi}\int_0^{\infty}\frac{x^{2026}+1}{(1+x^{2026})(1+x^2)}\,dx$$
$$=\frac{1}{\pi}\int_0^{\infty}\frac{1}{1+x^2}\,dx.$$

Since

$$\int_0^{\infty}\frac{dx}{1+x^2}=\left[\tan^{-1}x\right]_0^{\infty}=\frac{\pi}{2},$$

we obtain

$$2I=\frac{1}{\pi}\cdot\frac{\pi}{2}=\frac{1}{2}.$$

Hence,

$$I=\frac{1}{4}.$$

Therefore,

$$\boxed{I=0.25.}$$

Solution for Q64

Q.64	Let X and Y be two real-valued random variables with E(X) = 1, E(Y) = 2, E(X²) = 4, E(Y²) = 9, and E(XY) = 0.9, where E denotes the expectation operator. The value of α that minimizes E((X−αY)²) is ________ (*Round off to one decimal place*)

Solution:
Given,

$$E(X)=1,\qquad E(Y)=2,\qquad E(X^2)=4,\qquad E(Y^2)=9,$$

We need to minimize

$$E\left[(X-\alpha Y)^2\right].$$

Expanding the square,

$$(X-\alpha Y)^2=X^2-2\alpha XY+\alpha^2Y^2.$$

Taking expectation on both sides,

$$E\left[(X-\alpha Y)^2\right]=E(X^2)-2\alpha E(XY)+\alpha^2E(Y^2).$$

Rahimul Islam Mazumdar

Substituting the given values,

$$E\left[(X-\alpha Y)^2\right] = 4-2(0.9)\alpha+9\alpha^2 = 4-1.8\alpha+9\alpha^2.$$

Differentiating with respect to α,

$$\frac{d}{d\alpha}E\left[(X-\alpha Y)^2\right] = 18\alpha-1.8.$$

Setting the derivative equal to zero,

$$18\alpha-1.8=0.$$

Hence,

$$\alpha=\frac{1.8}{18}=0.1.$$

Since

$$\frac{d^2}{d\alpha^2}E\left[(X-\alpha Y)^2\right]=18>0,$$

the value corresponds to a minimum.
Therefore,

$$\boxed{\alpha=0.1.}$$

Solution for Q63

Q.63	A uniform spherical volume charge distribution of radius 2 m, centered at the origin, has a strength of $\frac{3}{\pi}\times 10^{-6}$ C/m^3. A point charge of strength $\pi \times 8.854\times 10^{-12}$ C is moved from (– 3, 0, – 4) to (0, 0, 4) in Cartesian coordinate system. The relative permittivity of the medium is 1 and the coordinate values are in meters. The work done during the process is _______ μJ *(Round off to two decimal places)*

Solution:
Given,

$$R=2\text{ m},\qquad \rho=\frac{3}{\pi}\times 10^{-6}\text{ C/m}^3,$$

and the point charge

$$q=\pi\times 8.854\times 10^{-12}=\pi\epsilon_0\text{ C},$$

where

$$\epsilon_0=8.854\times 10^{-12}\text{ F/m}.$$

The point charge is moved from

$$A(-3,0,-4)$$

to

$$B(0,0,4).$$

The distances of the two points from the origin are

$$r_A=\sqrt{(-3)^2+(-4)^2}=5\text{ m},$$

and

$$r_B=\sqrt{0^2+0^2+4^2}=4\text{ m}.$$

Since both r_A and r_B are greater than the sphere radius $R=2$ m, both points lie outside the charged sphere.
The total charge enclosed in the sphere is

$$Q=\rho\left(\frac{4}{3}\pi R^3\right).$$

Substituting the given values,

$$Q=\frac{3}{\pi}\times 10^{-6}\times\frac{4}{3}\pi(2)^3=32\times 10^{-6}=3.2\times 10^{-5}\text{ C}.$$

The electric potential outside a uniformly charged sphere is

$$V(r)=\frac{1}{4\pi\epsilon_0}\frac{Q}{r}.$$

Hence,

$$V_A=\frac{1}{4\pi\epsilon_0}\frac{Q}{5},$$

and

$$V_B=\frac{1}{4\pi\epsilon_0}\frac{Q}{4}.$$

Therefore,

$$V_B-V_A=\frac{Q}{4\pi\epsilon_0}\left(\frac{1}{4}-\frac{1}{5}\right)=\frac{Q}{80\pi\epsilon_0}.$$

The work done in moving the charge is

$$W=q(V_B-V_A).$$

Substituting $q=\pi\epsilon_0$,

$$W=\pi\epsilon_0\cdot\frac{Q}{80\pi\epsilon_0}=\frac{Q}{80}.$$

Thus,

$$W=\frac{3.2\times 10^{-5}}{80}=4\times 10^{-7}\text{ J}=0.40\ \mu\text{J}.$$

Therefore,

$$\boxed{W=0.40\ \mu\text{J}.}$$

Rahimul Islam Mazumdar

Solution for Q62

Q.62 The magnitude of the contour integral

$\oint_C \left(\frac{(z+1)^2}{(z-i)(z-2)}\right) dz$

over the contour $C: |z - 2 - i| = 3/2$ is ______

(*Round off to two decimal places*)

Note: z is a complex variable and $i = \sqrt{-1}$

Solution:
The given contour integral is

$$I = \oint_C \frac{(z+1)^2}{(z-i)(z-2)}\, dz,$$

where the contour is

$$|z - 2 - i| = \frac{3}{2}.$$

The integrand has simple poles at

$$z = i \quad \text{and} \quad z = 2.$$

The contour is a circle centered at

$$2 + i$$

with radius

$$\frac{3}{2}.$$

The distances of the poles from the center are

$$|\, i - (2+i)\,| = 2 > \frac{3}{2},$$

and

$$|\, 2 - (2+i)\,| = 1 < \frac{3}{2}.$$

Hence, the pole at $z = i$ lies outside the contour, while the pole at $z = 2$ lies inside the contour.

By Cauchy's Residue Theorem,

$$I = 2\pi i \ \text{Res}\left(\frac{(z+1)^2}{(z-i)(z-2)},\ z = 2\right).$$

The residue at $z = 2$ is

$$\text{Res} = \lim_{z \to 2} (z-2) \frac{(z+1)^2}{(z-i)(z-2)} = \frac{(2+1)^2}{2-i} = \frac{9}{2-i}.$$

Therefore,

$$I = 2\pi i \cdot \frac{9}{2-i} = \frac{18\pi i}{2-i}.$$

The magnitude of the contour integral is

$$|I| = \frac{18\pi |i|}{|2-i|} = \frac{18\pi}{\sqrt{2^2 + (-1)^2}} = \frac{18\pi}{\sqrt{5}}.$$

Numerically,

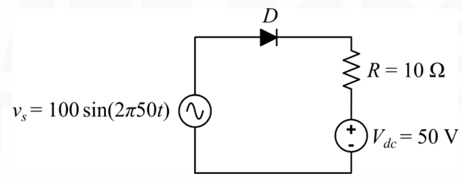
$$|I| = \frac{18\pi}{\sqrt{5}} \approx 25.29.$$

Therefore,

$$\boxed{|I| = 25.29.}$$

Solution for Q61

Q.61 Consider the circuit shown. Assume that the diode (D) is ideal.



Given $v_s = 100 \sin(2\pi 50t)$ V, $V_{dc} = 50$ V, and $R = 10\ \Omega$, the average value of the current through the diode is __________ A

(*Round off to two decimal places*)

Solution:
The source voltage is

$$v_s = 100 \sin(\omega t),$$

where

$$\omega = 2\pi(50) \text{ rad/s}.$$

Since the diode is ideal, it conducts only when it is forward biased.

Applying KVL,

$$v_s = V_{dc} + iR.$$

Hence, the diode current is

$$i = \frac{v_s - V_{dc}}{R} = \frac{100 \sin\theta - 50}{10},$$

where

$$\theta = \omega t.$$

The diode conducts only when

$$100 \sin\theta > 50,$$

Rahimul Islam Mazumdar

or

$$\sin\theta > \frac{1}{2}.$$

Thus, the conduction interval is

$$\frac{\pi}{6} \le \theta \le \frac{5\pi}{6}.$$

Outside this interval,

$$i = 0.$$

The average current over one cycle is

$$I_{\text{avg}} = \frac{1}{2\pi}\int_{\pi/6}^{5\pi/6} \frac{100\sin\theta - 50}{10}\, d\theta.$$

Simplifying,

$$I_{\text{avg}} = \frac{1}{2\pi}\int_{\pi/6}^{5\pi/6} (10\sin\theta - 5)\, d\theta.$$

Evaluating the integral,

$$I_{\text{avg}} = \frac{1}{2\pi}\left[-10\cos\theta - 5\theta\right]_{\pi/6}^{5\pi/6}.$$

Using

$$\cos\frac{5\pi}{6} = -\frac{\sqrt{3}}{2}, \qquad \cos\frac{\pi}{6} = \frac{\sqrt{3}}{2},$$

we obtain

$$I_{\text{avg}} = \frac{1}{2\pi}\left(10\sqrt{3} - \frac{10\pi}{3}\right).$$

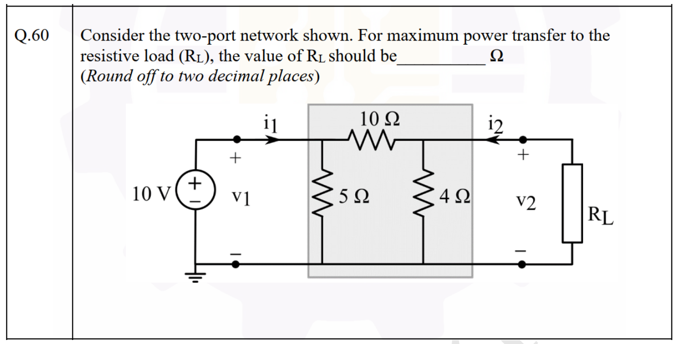
Substituting the numerical values,

$$I_{\text{avg}} = \frac{17.3205 - 10.4720}{6.2832} \approx 1.09 \text{ A}.$$

Therefore,

$$\boxed{I_{\text{avg}} = 1.09 \text{ A}.}$$

Solution for Q60


Q.60 Consider the two-port network shown. For maximum power transfer to the resistive load (R_L), the value of R_L should be__________ Ω

(Round off to two decimal places)

Solution:

For maximum power transfer,

$$R_L = R_{\text{th}},$$

where R_{th} is the Thevenin resistance seen from the load terminals.

To find R_{th}, deactivate the independent voltage source. Since it is an ideal voltage source, replace it with a short circuit.

This shorts the left node to ground. Hence, the $5\,\Omega$ resistor is bypassed, and the network seen from the load terminals consists of

$$4\ \Omega$$

in parallel with

$$10\ \Omega.$$

Therefore,

$$R_{\text{th}} = 4 \parallel 10 = \frac{4\times 10}{4+10} = \frac{40}{14} = \frac{20}{7} = 2.857\ \Omega.$$

Hence,

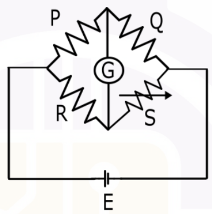
$$R_L = R_{\text{th}} = 2.857\ \Omega.$$

Therefore,

$$\boxed{R_L = 2.86\ \Omega.}$$

Rahimul Islam Mazumdar

Solution for Q57

Q.57	The resistance values of the Wheatstone bridge shown are: $P = 2000\ \Omega$, $Q = 500\ \Omega$, $R = 3000\ \Omega$. The battery voltage $E = 50\,V$. The battery has an internal resistance of $1\ \Omega$ and the Galvanometer (G) has a resistance of $50\ \Omega$.  The value of the resistance S for balanced condition is ____________ Ω (*Answer in integer*)

Solution:
For a balanced Wheatstone bridge,

$$\frac{P}{Q} = \frac{R}{S}.$$

The battery voltage, internal resistance, and galvanometer resistance do not affect the balance condition since the galvanometer current is zero at balance.

Hence,

$$S = \frac{RQ}{P}.$$

Substituting the given values,

$$S = \frac{3000 \times 500}{2000} = 750\ \Omega.$$

Therefore,

$$\boxed{S = 750\ \Omega.}$$

Solution for Q56

Q.56	Consider the system of linear equations: $\boldsymbol{Ax} = \boldsymbol{b}$, where A is an $n \times n$ matrix, and $\boldsymbol{x}$ and $\boldsymbol{b}$ are n-dimensional column vectors. Suppose this system of equations has a unique solution. Which of the following statements is/are correct?
(A)	A^{-1} exists
(B)	The system of equations $A^m\boldsymbol{x} = \boldsymbol{b}$ also has a unique solution for $m = 1, 2, 3, \ldots$
(C)	$\text{rank}(A) = \text{rank}(A^m)$, for $m = 1, 2, 3, \ldots$
(D)	$\text{rank}(A) < \text{rank}([A \mid \boldsymbol{b}\,])$, where $[A \mid \boldsymbol{b}\,]$ denotes the augmented matrix.

Solution:
Since the system

$$Ax = b$$

has a unique solution, the matrix A must be invertible. Therefore,

$$\det(A) \neq 0,$$

or equivalently,

$$\text{rank}(A) = n.$$

Now examine each statement.

(A) Since A is invertible,

$$A^{-1}$$

exists.
Hence, (A) is true.

(B) Since A is invertible,

$$A^m = A \cdot A \cdots A$$

is also invertible for every positive integer m. Therefore,

$$A^m x = b$$

has the unique solution

$$x = (A^m)^{-1} b.$$

Hence, (B) is true.

(C) Since A is invertible,

$$\text{rank}(A) = n.$$

Also, A^m is invertible for every positive integer m, so

$$\text{rank}(A^m) = n.$$

Thus,

$$\text{rank}(A) = \text{rank}(A^m).$$

Hence, (C) is true.

(D) A system has a unique solution only when

$$\text{rank}(A) = \text{rank}([A \mid b]) = n.$$

Rahimul Islam Mazumdar

Therefore,

$$\text{rank}(A) < \text{rank}([A \mid b])$$

is false.
Hence, (D) is false.
Therefore,

$$\boxed{\text{(A), (B), and (C) are correct.}}$$

SOLUTION FOR Q53

Q.53	A system is represented in state-space form as follows: (u: input, $\mathbf{x}$: state vector, y: output) $$\dot{\mathbf{x}} = \begin{bmatrix} 1 & 2 \\ -3 & 0 \end{bmatrix}\mathbf{x} + \begin{bmatrix} 1 \\ 2 \end{bmatrix}u$$ $$y = [1 \quad 2]\,\mathbf{x}$$ Consider the new state vector $\mathbf{z} = \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix}\mathbf{x}$ What is the state-space representation of the system in terms of the new state vector $\mathbf{z}$?
(A)	$\dot{\mathbf{z}} = \begin{bmatrix} -1 & 4 \\ -1 & -2 \end{bmatrix}\mathbf{z} + \begin{bmatrix} 4 \\ -1 \end{bmatrix}u$ $y = [2 \quad 3]\,\mathbf{z}$
(B)	$\dot{\mathbf{z}} = \begin{bmatrix} 2 & 3 \\ 0 & 3 \end{bmatrix}\mathbf{z} + \begin{bmatrix} 3 \\ 5 \end{bmatrix}u$ $y = [2 \quad 3]\,\mathbf{z}$
(C)	$\dot{\mathbf{z}} = \begin{bmatrix} 4 & 9 \\ -2 & -3 \end{bmatrix}\mathbf{z} + \begin{bmatrix} 4 \\ -1 \end{bmatrix}u$ $y = [2 \quad 3]\,\mathbf{z}$
(D)	$\dot{\mathbf{z}} = \begin{bmatrix} 2 & 1 \\ -4 & 1 \end{bmatrix}\mathbf{z} + \begin{bmatrix} 4 \\ -1 \end{bmatrix}u$ $y = [4 \quad -1]\,\mathbf{z}$

Solution:
Given,

$$A = \begin{bmatrix} 1 & 2 \\ -3 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad C = \begin{bmatrix} 1 & 2 \end{bmatrix},$$

and the state transformation

$$z = Tx,$$

where

$$T = \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix}.$$

The transformed matrices are

$$A_z = TAT^{-1}, \qquad B_z = TB, \qquad C_z = CT^{-1}.$$

The determinant of T is

$$\det(T) = 2(0) - 1(-1) = 1,$$

hence,

$$T^{-1} = \begin{bmatrix} 0 & -1 \\ 1 & 2 \end{bmatrix}.$$

Now,

$$AT^{-1} = \begin{bmatrix} 1 & 2 \\ -3 & 0 \end{bmatrix}\begin{bmatrix} 0 & -1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 0 & 3 \end{bmatrix}.$$

Therefore,

$$A_z = TAT^{-1} = \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix}\begin{bmatrix} 2 & 3 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 4 & 9 \\ -2 & -3 \end{bmatrix}.$$

Also,

$$B_z = TB = \begin{bmatrix} 2 & 1 \\ -1 & 0 \end{bmatrix}\begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ -1 \end{bmatrix}.$$

Finally,

$$C_z = CT^{-1} = \begin{bmatrix} 1 & 2 \end{bmatrix}\begin{bmatrix} 0 & -1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 3 \end{bmatrix}.$$

Hence, the transformed state-space model is

$$\dot{z} = \begin{bmatrix} 4 & 9 \\ -2 & -3 \end{bmatrix} z + \begin{bmatrix} 4 \\ -1 \end{bmatrix} u,$$

$$y = \begin{bmatrix} 2 & 3 \end{bmatrix} z.$$

Therefore,

$$\boxed{\text{Option (C).}}$$

Rahimul Islam Mazumdar
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SOLUTION FOR Q52

Q.52	A system is characterized by the following state equation and output equation (u: input, $\mathbf{x}$: state vector, y: output) $$\dot{\mathbf{x}} = \begin{bmatrix} a & b \\ -a & 0 \end{bmatrix}\mathbf{x} + \begin{bmatrix} 1 \\ 0 \end{bmatrix}u$$ $$y = [1 \quad 2]\,\mathbf{x}$$ What are the values of a and b for which the poles of the transfer function are at $-2 + j3$ and $-2 - j3$?
(A)	$a = 4,\ b = 3.25$
(B)	$a = -4,\ b = 3.25$
(C)	$a = 4,\ b = -3.25$
(D)	$a = -4,\ b = -3.25$

Solution:
Given,

$$A = \begin{bmatrix} a & b \\ -a & 0 \end{bmatrix}.$$

The poles of the transfer function are the eigenvalues of the state matrix A.
The characteristic equation is

$$\det(sI - A) = 0.$$

Now,

$$sI - A = \begin{bmatrix} s-a & -b \\ a & s \end{bmatrix}.$$

Hence,

$$\det(sI - A) = (s-a)s - (-b)a = s^2 - as + ab.$$

Therefore, the characteristic equation is

$$s^2 - as + ab = 0.$$

The desired poles are

$$-2 \pm j3.$$

Hence, the desired characteristic equation is

$$(s + 2 - j3)(s + 2 + j3) = 0,$$

or

$$(s+2)^2 + 3^2 = 0,$$

which simplifies to

$$s^2 + 4s + 13 = 0.$$

Comparing

$$s^2 - as + ab = s^2 + 4s + 13,$$

gives

$$-a = 4 \quad \Rightarrow \quad a = -4,$$

and

$$ab = 13.$$

Therefore,

$$(-4)b = 13,$$

which gives

$$b = -\frac{13}{4} = -3.25.$$

Hence,

$$\boxed{a = -4, \qquad b = -3.25.}$$

Therefore,

$$\boxed{\text{Option (D).}}$$

SOLUTION FOR Q51

Solution:
The loop lies in the XY-plane, so the area vector is along the z-axis. Therefore, only the z-components of the magnetic fields contribute to the magnetic flux.
For region a_1,

$$\mathbf{B}_1 = \frac{m}{\sqrt{2}}\sin(\omega t)(\hat{x} + 0\hat{y} + \hat{z}),$$

so that

$$B_{1z} = \frac{m}{\sqrt{2}}\sin(\omega t).$$

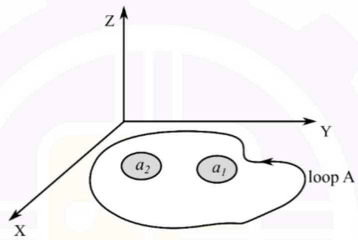
For region a_2,

$$\mathbf{B}_2 = -\frac{n}{\sqrt{2}}\cos\left(2\omega t + \frac{\pi}{4}\right)(0\hat{x} + \hat{y} + \hat{z}),$$

hence,

$$B_{2z} = -\frac{n}{\sqrt{2}}\cos\left(2\omega t + \frac{\pi}{4}\right).$$

Rahimul Islam Mazumdar

Q.51	The figure shows an arbitrarily shaped planar conducting loop A in the XY plane. Two nonintersecting regions with areas a_1 and a_2 within the loop are subjected to magnetic fields $\vec{B}_1 = \frac{m}{\sqrt{2}}\sin(\omega t)\,(1\,\hat{x} + 0\,\hat{y} + 1\,\hat{z})$, and $\vec{B}_2 = -\frac{n}{\sqrt{2}}\cos(2\omega t + \pi/4)\,(0\,\hat{x} + 1\,\hat{y} + 1\,\hat{z})$, respectively.  What is the expression for the induced rms voltage in loop A?
(A)	$\sqrt{\frac{a_1^2\omega^2m^2 + 4a_2^2\omega^2n^2}{4}}$
(B)	$\sqrt{\frac{a_1^2\omega^2m^2 + 4a_2^2\omega^2n^2}{2}}$
(C)	$\sqrt{\frac{a_1^2\omega^2m^2 - 2a_2^2\omega^2n^2}{2}}$
(D)	$\sqrt{a_1^2\omega^2m^2 + 2a_2^2\omega^2n^2}$

The total magnetic flux through the loop is

$$\Phi(t) = a_1B_{1z} + a_2B_{2z},$$

or

$$\Phi(t) = \frac{a_1m}{\sqrt{2}}\sin(\omega t) - \frac{a_2n}{\sqrt{2}}\cos\left(2\omega t + \frac{\pi}{4}\right).$$

Using Faraday's law,

$$e(t) = -\frac{d\Phi}{dt},$$

which gives

$$e(t) = -\frac{a_1m\omega}{\sqrt{2}}\cos(\omega t) - \frac{2a_2n\omega}{\sqrt{2}}\sin\left(2\omega t + \frac{\pi}{4}\right).$$

Since the two components are at different frequencies, their RMS values combine as

$$E_{\text{rms}} = \sqrt{E_{1,\text{rms}}^2 + E_{2,\text{rms}}^2}.$$

Now,

$$E_{1,\text{rms}} = \frac{a_1m\omega}{2},$$

and

$$E_{2,\text{rms}} = a_2n\omega.$$

Hence,

$$E_{\text{rms}} = \sqrt{\left(\frac{a_1m\omega}{2}\right)^2 + (a_2n\omega)^2},$$

or

$$E_{\text{rms}} = \sqrt{\frac{a_1^2\omega^2m^2 + 4a_2^2\omega^2n^2}{4}}.$$

Therefore,

$$\boxed{E_{\text{rms}} = \sqrt{\frac{a_1^2\omega^2m^2 + 4a_2^2\omega^2n^2}{4}}}$$

Hence,

$$\boxed{\text{Option (A).}}$$

Solution for Q50

Q.50	Consider the second-order differential equation $$\frac{d^2y}{dx^2} + \frac{dy}{dx} + y = 0$$ with initial conditions $$y(0) = 1, \qquad \left.\frac{dy}{dx}\right\|_{x=0} = 1$$ The solution is given by
(A)	$y(x) = \exp(-x/2)\left(\cos\left(\frac{\sqrt{3}x}{2}\right) + \sqrt{3}\sin\left(\frac{\sqrt{3}x}{2}\right)\right)$
(B)	$y(x) = \exp(-x/2)\left(\cos\left(\frac{\sqrt{3}x}{2}\right) + \frac{1}{\sqrt{3}}\sin\left(\frac{\sqrt{3}x}{2}\right)\right)$
(C)	$y(x) = \exp(-x/2)\left(\cos\left(\frac{\sqrt{3}x}{2}\right) - \frac{1}{\sqrt{3}}\sin\left(\frac{\sqrt{3}x}{2}\right)\right)$
(D)	$y(x) = \exp(-x/2)\left(\cos\left(\frac{\sqrt{3}x}{2}\right) - \sqrt{3}\sin\left(\frac{\sqrt{3}x}{2}\right)\right)$

Solution:
The given differential equation is

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} + y = 0,$$

with initial conditions

$$y(0) = 1, \qquad y'(0) = 1.$$

Assume a solution of the form

Rahimul Islam Mazumdar

$$y = e^{rx}.$$

The characteristic equation is

$$r^2 + r + 1 = 0.$$

Using the quadratic formula,

$$r = \frac{-1 \pm \sqrt{1-4}}{2} = -\frac{1}{2} \pm j\frac{\sqrt{3}}{2}.$$

Hence, the general solution is

$$y(x) = e^{-x/2}\left(C_1 \cos\frac{\sqrt{3}x}{2} + C_2 \sin\frac{\sqrt{3}x}{2}\right).$$

Using the initial condition

$$y(0) = 1,$$

gives

$$C_1 = 1.$$

Thus,

$$y(x) = e^{-x/2}\left(\cos\frac{\sqrt{3}x}{2} + C_2 \sin\frac{\sqrt{3}x}{2}\right).$$

Differentiating,

$$\begin{aligned} y'(x) = &-\frac{1}{2}e^{-x/2}\left(\cos\frac{\sqrt{3}x}{2} + C_2 \sin\frac{\sqrt{3}x}{2}\right) \\ &+ e^{-x/2}\left(-\frac{\sqrt{3}}{2}\sin\frac{\sqrt{3}x}{2} + C_2\frac{\sqrt{3}}{2}\cos\frac{\sqrt{3}x}{2}\right). \end{aligned}$$

Applying the initial condition

$$y'(0) = 1,$$

yields

$$-\frac{1}{2} + \frac{\sqrt{3}}{2}C_2 = 1.$$

Therefore,

$$\sqrt{3}\,C_2 = 3,$$

or

$$C_2 = \sqrt{3}.$$

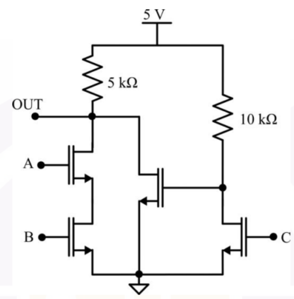
Hence,

$$\boxed{y(x) = e^{-x/2}\left(\cos\frac{\sqrt{3}x}{2} + \sqrt{3}\sin\frac{\sqrt{3}x}{2}\right).}$$

Therefore,

$$\boxed{\text{Option (A).}}$$

SOLUTION FOR Q46

Q.46	The MOSFET switches shown in the circuit are ideal.  Which of the following is the correct option for Boolean logical expression of the output (OUT), and the maximum possible power (P) consumed by the circuit?
(A)	OUT = $\overline{AB + \overline{C}}$, P = 5 mW
(B)	OUT = $\overline{(A + B)\overline{C}}$, P = 5 mW
(C)	OUT = $\overline{AB\overline{C}} + \overline{C}$, P = 7.5 mW
(D)	OUT = $\overline{AB}C$, P = 7.5 mW

Step 1: Determine the control signal generated by input C

The node connected to the 10 $k\Omega$ resistor is pulled up to 5 V. An NMOS transistor controlled by C can pull this node to ground.

Therefore,

$$X = \overline{C}$$

Thus, the middle NMOS transistor conducts when

$$\overline{C} = 1.$$

Step 2: Determine the pull-down condition at the output

The output node is connected to ground through two possible paths:

1) Two NMOS transistors in series controlled by A and B, which conduct when

$$AB = 1.$$

Rahimul Islam

2) A single NMOS transistor controlled by $\overline{C}$, which conducts when
$$\overline{C} = 1.$$

Since the two paths are in parallel, the pull-down condition is

$$PD = AB + \overline{C}.$$

Applying De Morgan's theorem,

$$PD = \overline{\overline{AB}\,C}.$$

Hence, the corresponding logic function is

$$\boxed{\text{OUT} = \overline{AB}\,C}.$$

Step 3: Determine the maximum power consumption

Maximum static power occurs when both branches draw current simultaneously.

Current through the 5 $k\Omega$ branch:

$$I_1 = \frac{5}{5k} = 1 \text{ mA}$$

$$P_1 = VI = 5 \times 1 \text{ mA} = 5 \text{ mW}$$

Current through the 10 $k\Omega$ branch:

$$I_2 = \frac{5}{10k} = 0.5 \text{ mA}$$

$$P_2 = VI = 5 \times 0.5 \text{ mA} = 2.5 \text{ mW}$$

Therefore,

$$P_{\text{max}} = P_1 + P_2$$

$$P_{\text{max}} = 5 + 2.5$$

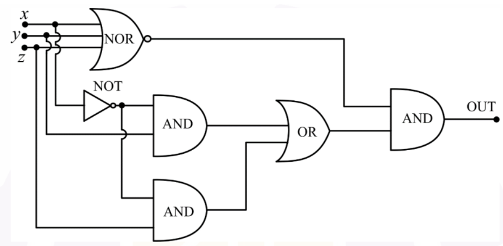
$$\boxed{P_{\text{max}} = 7.5 \text{ mW}}$$

Hence,

$$\boxed{\text{OUT} = \overline{AB}\,C, \qquad P_{\text{max}} = 7.5 \text{ mW}}$$

$$\boxed{\text{Option (D)}}$$

Solution for Q45

Q.45	The digital circuit shown has 3 inputs (x, y, and z).  The simplified logical expression for the output (OUT) is:
(A)	$\bar{x}\,\overline{yz}$
(B)	0
(C)	$\bar{x}(y + z)$
(D)	1

NOR gate output:

$$N = \overline{x + y + z}$$

Upper AND gate:
The NOT gate produces

$$\bar{x}$$

Hence,

$$A_1 = \bar{x}y$$

Lower AND gate:

$$A_2 = x\bar{x} = 0$$

OR gate output:

$$O = A_1 + A_2$$

$$O = \bar{x}y + 0$$

$$O = \bar{x}y$$

Final output:

$$\text{OUT} = \overline{x + y + z} \cdot (\bar{x}y)$$

Applying De Morgan's law,

$$\overline{x + y + z} = \bar{x}\bar{y}\bar{z}$$

Thus,

$$\text{OUT} = (\bar{x}\bar{y}\bar{z})(\bar{x}y)$$

$$= \bar{x}\bar{x}\bar{y}y\bar{z}$$

Since

$$\bar{y}y = 0,$$

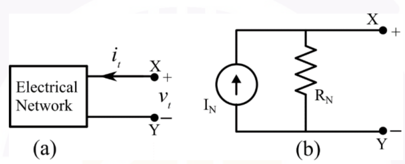
$$\boxed{\text{OUT} = 0}$$

$$\boxed{\text{Option (B)}}$$

SOLUTION FOR Q44

Q.44 The terminal voltage and current of a linear electrical network shown in Figure (a) are given in the table.

Terminal voltage (v_t)	Terminal current (i_t)
18 V	– 0.5 A
30 V	0.5 A
36 V	1.0 A



The correct choice for the parameters (I_N, R_N) of the Norton equivalent circuit shown in Figure (b) is:

(A)	$I_N = 3.0$ A, $R_N = 24.0\ \Omega$
(B)	$I_N = 12.0$ A, $R_N = 2.0\ \Omega$
(C)	$I_N = 2.0$ A, $R_N = 12.0\ \Omega$
(D)	$I_N = 2.0$ A, $R_N = 24.0\ \Omega$

Given data:

v_t (V)	i_t (A)
18	−0.5
30	0.5
36	1.0

Since the network is linear, the terminal voltage-current relationship is a straight line.

Step 1: Determine the Norton resistance
Using any two points,

$$R_N = \frac{\Delta v_t}{\Delta i_t}$$

Using $(18, -0.5)$ and $(30, 0.5)$,

$$R_N = \frac{30 - 18}{0.5 - (-0.5)}$$

$$= \frac{12}{1}$$

$$\boxed{R_N = 12\ \Omega}$$

Step 2: Determine the open-circuit voltage
The v_t-i_t relation can be written as

$$v_t = V_{OC} + i_t R_N$$

Substituting $R_N = 12\ \Omega$ and the point $(v_t, i_t) = (30, 0.5)$,

$$30 = V_{OC} + 0.5(12)$$

$$30 = V_{OC} + 6$$

$$V_{OC} = 24\ \text{V}$$

$$\boxed{V_{OC} = 24\ \text{V}}$$

Step 3: Determine the Norton current
The Norton current is

$$I_N = \frac{V_{OC}}{R_N}$$

$$I_N = \frac{24}{12}$$

$$\boxed{I_N = 2\ \text{A}}$$

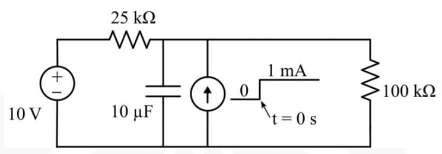
Therefore,

$$\boxed{I_N = 2.0\ \text{A}, \qquad R_N = 12.0\ \Omega}$$

$$\boxed{\text{Option (C)}}$$

Rahimul Islam Mazumdar

SOLUTION FOR Q43

Q.43	The electrical network shown has an independent voltage source (10 V) and a current source (1 u(t) mA).  The voltage across the capacitor at time instants (in seconds) $t = 0^+$, t = 0.50, and $t = \infty$, respectively, is:
(A)	8.00 V, 28.00 V, 26.36 V
(B)	8.00 V, 26.36 V, 28.00 V
(C)	10.00 V, 26.36 V, 28.00 V
(D)	10.00 V, 28.00 V, 26.36 V

Given:

- Voltage source: 10 V
- Current source: $1u(t)$ mA
- Resistors: 25 $k\Omega$ and 100 $k\Omega$
- Capacitor: 10 μF

Let the capacitor voltage be denoted by $v_C(t)$.

Step 1: Determine $v_C(0^-)$

Before $t = 0$, the current source is OFF (0 mA). At steady state, the capacitor behaves as an open circuit.

Using voltage division,

$$v_C(0^-) = 10\left(\frac{100k}{25k + 100k}\right)$$

$$= 10\left(\frac{100}{125}\right)$$

$$= 8 \text{ V}$$

Since the voltage across a capacitor cannot change instantaneously,

$$\boxed{v_C(0^+) = v_C(0^-) = 8 \text{ V}}$$

Step 2: Determine $v_C(\infty)$

For $t > 0$, the current source supplies

$$I_s = 1 \text{ mA}$$

At steady state, the capacitor again behaves as an open circuit.

Applying KCL at the top node,

$$\frac{V - 10}{25k} + \frac{V}{100k} = 1 \text{ mA}$$

Multiplying by $100k$,

$$4(V - 10) + V = 100$$

$$5V - 40 = 100$$

$$5V = 140$$

$$V = 28 \text{ V}$$

Therefore,

$$\boxed{v_C(\infty) = 28 \text{ V}}$$

Step 3: Determine the time constant

To find the equivalent resistance seen by the capacitor, deactivate the independent sources:

- Voltage source $\rightarrow$ short circuit
- Current source $\rightarrow$ open circuit

Thus,

$$R_{th} = 25k\Omega \parallel 100k\Omega$$

$$= \frac{25 \times 100}{25 + 100} k\Omega$$

$$= 20k\Omega$$

Hence,

$$\tau = R_{th}C$$

$$= (20k)(10\mu F)$$

$$= 0.2 \text{ s}$$

Step 4: Determine $v_C(0.50)$

For a first-order RC circuit,

$$v_C(t) = v_C(\infty) + \left[v_C(0^+) - v_C(\infty)\right] e^{-t/\tau}$$

Substituting the values,

$$v_C(0.50) = 28 + (8 - 28)e^{-0.50/0.20}$$

$$= 28 - 20e^{-2.5}$$

Rahimul Islam Mazumdar

Using

$$e^{-2.5} = 0.0821,$$

$$v_C(0.50) = 28 - 20(0.0821)$$

$$= 28 - 1.642$$

$$= 26.36 \text{ V}$$

$$\boxed{v_C(0.50) = 26.36 \text{ V}}$$

Therefore,

$$v_C(0^+) = 8.00 \text{ V}$$

$$v_C(0.50) = 26.36 \text{ V}$$

$$v_C(\infty) = 28.00 \text{ V}$$

$$\boxed{(8.00 \text{ V},\ 26.36 \text{ V},\ 28.00 \text{ V})}$$

$$\boxed{\text{Option (B)}}$$

SOLUTION FOR Q42

Q.42	An electrical component has voltage drop $v = V_m \sin(\omega t)$, when the current through it is $i = I_m \sin(\omega t - \theta)$. What is the average power dissipated over a half cycle corresponding to ω?
(A)	0
(B)	$V_m I_m \cos\theta$
(C)	$\frac{V_m I_m}{2} \cos\theta$
(D)	$\frac{V_m I_m}{4} \cos\theta$

Given:

$$v(t) = V_m \sin(\omega t)$$

$$i(t) = I_m \sin(\omega t - \theta)$$

The instantaneous power is

$$p(t) = v(t)i(t)$$

$$p(t) = V_m I_m \sin(\omega t) \sin(\omega t - \theta)$$

Using the trigonometric identity

$$\sin A \sin B = \frac{1}{2}\left[\cos(A - B) - \cos(A + B)\right],$$

we obtain

$$p(t) = \frac{V_m I_m}{2}\left[\cos\theta - \cos(2\omega t - \theta)\right].$$

Average power over a half cycle

The half-cycle duration is

$$T_h = \frac{\pi}{\omega}.$$

Hence,

$$P_{\text{avg}} = \frac{1}{T_h}\int_0^{T_h} p(t)\,dt.$$

Substituting $p(t)$,

$$P_{\text{avg}} = \frac{V_m I_m}{2}\left[\cos\theta - \frac{1}{T_h}\int_0^{T_h} \cos(2\omega t - \theta)\,dt\right].$$

Since $\cos(2\omega t - \theta)$ completes one full cycle over the interval $0 \le t \le \pi/\omega$,

$$\int_0^{T_h} \cos(2\omega t - \theta)\,dt = 0.$$

Therefore,

$$P_{\text{avg}} = \frac{V_m I_m}{2}\cos\theta.$$

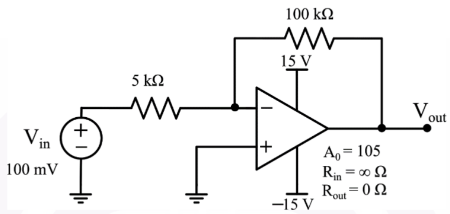
$$\boxed{P_{\text{avg}} = \frac{V_m I_m}{2}\cos\theta}$$

Hence,

$$\boxed{\text{Option (C)}}$$

Rahimul Islam Mazumdar

Solution for Q37

Q.37	In the circuit shown, the open loop gain of the operational amplifier is $A_0 = 105$.  What is the voltage gain of the circuit? *(Round off to two decimal places)*
(A)	–16.67
(B)	–20.00
(C)	–21.00
(D)	–12.67

Given:

$$A_0 = 105, \qquad R_1 = 5\ k\Omega, \qquad R_f = 100\ k\Omega$$

The non-inverting terminal is grounded, hence

$$V_+ = 0$$

For a non-ideal op-amp,

$$V_o = A_0(V_+ - V_-)$$

Substituting $V_+ = 0$,

$$V_o = -105V_-$$

Therefore,

$$V_- = -\frac{V_o}{105}$$

Applying KCL at the inverting node: Since the input current to the op-amp is zero,

$$\frac{V_{in} - V_-}{R_1} = \frac{V_- - V_o}{R_f}$$

Substituting $R_1 = 5\ k\Omega$ and $R_f = 100\ k\Omega$,

$$\frac{V_{in} - V_-}{5} = \frac{V_- - V_o}{100}$$

Multiplying both sides by 100,

$$20(V_{in} - V_-) = V_- - V_o$$

Expanding,

$$20V_{in} - 20V_- = V_- - V_o$$

$$20V_{in} = 21V_- - V_o$$

Substituting

$$V_- = -\frac{V_o}{105},$$

gives

$$20V_{in} = 21\left(-\frac{V_o}{105}\right) - V_o$$

$$20V_{in} = -\frac{21}{105}V_o - V_o$$

$$20V_{in} = -\left(\frac{1}{5} + 1\right)V_o$$

$$20V_{in} = -\frac{6}{5}V_o$$

Hence,

$$V_o = -\frac{100}{6}V_{in}$$

Therefore, the voltage gain is

$$A_v = \frac{V_o}{V_{in}} = -\frac{100}{6} = -16.67$$

$$\boxed{A_v = -16.67}$$

Thus, the correct answer is

$$\boxed{\text{Option (A)}}$$

Solution for Q26

Q.26	A single-phase voltage source $v_s = 325\sin(2\pi 50t)$ V delivers a current, $i = 12\sin(2\pi 50t) + 9\sin(2\pi 150t)$ A to a load. The load power factor, correct up to two decimal places, is:
(A)	1.00
(B)	0.80
(C)	0.65
(D)	0.57

Given:

$$v_s = 325\sin(2\pi 50t)\ \text{V}$$

$$i = 12\sin(2\pi 50t) + 9\sin(2\pi 150t)\ \text{A}$$

Step 1: Calculate the RMS voltage

$$V_{\text{rms}} = \frac{325}{\sqrt{2}} = 229.8\ \text{V}$$

Step 2: Calculate the RMS current

The current contains a fundamental component and a third-harmonic component.

Fundamental RMS current:

$$I_1 = \frac{12}{\sqrt{2}} = 8.485\ \text{A}$$

Third-harmonic RMS current:

$$I_3 = \frac{9}{\sqrt{2}} = 6.364\ \text{A}$$

Therefore, the total RMS current is

$$I_{\text{rms}} = \sqrt{I_1^2 + I_3^2}$$

$$= \sqrt{8.485^2 + 6.364^2}$$

$$= \sqrt{72 + 40.5}$$

$$= \sqrt{112.5}$$

$$= 10.606\ \text{A}$$

Step 3: Calculate the real power

Only voltage and current components of the same frequency contribute to the average power.

Since the source voltage contains only the fundamental component and it is in phase with the fundamental current,

$$P = V_{\text{rms}} I_1$$

$$= 229.8 \times 8.485$$

$$= 1950\ \text{W}$$

Step 4: Calculate the apparent power

$$S = V_{\text{rms}} I_{\text{rms}}$$

$$= 229.8 \times 10.606$$

$$= 2437.5\ \text{VA}$$

Step 5: Calculate the power factor

$$\text{Power Factor} = \frac{P}{S}$$

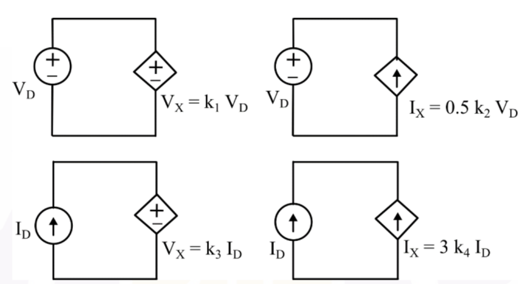
$$= \frac{1950}{2437.5}$$

$$= 0.80$$

$$\boxed{\text{Power Factor} = 0.80}$$

$$\boxed{\text{Option (B)}}$$

Solution for Q25

Q.25	Refer to the four circuits shown.  Which one of the following options for k_1, k_2, k_3, and k_4 makes all of them realizable?
(A)	$k_1 = 1$, $k_4 = -1/3$, for all values of k_2 and k_3
(B)	$k_2 = -2$, $k_3 = +1/3$, for all values of k_1 and k_4
(C)	$k_1 = 2$, $k_2 = 0.5$, $k_3 = -2/3$, $k_4 = -3$
(D)	$k_1 = 2$, $k_2 = -0.5$, $k_3 = -2/3$, $k_4 = +3$

Circuit 1: Voltage source and dependent voltage source

Since both voltage sources are connected across the same two nodes,

Rahimul Islam Mazumdar

$$V_X = V_D$$

Given

$$V_X = k_1 V_D$$

Hence,

$$k_1 = 1$$

Circuit 2: Voltage source and dependent current source

The voltage source fixes the voltage while the current source can supply any required current.

Therefore,

$$\boxed{\text{No restriction on } k_2}$$

Circuit 3: Current source and dependent voltage source

The current source fixes the loop current and the voltage source can assume any required voltage.

Therefore,

$$\boxed{\text{No restriction on } k_3}$$

Circuit 4: Current source and dependent current source

For realizability, the series current sources must impose the same loop current.

Given

$$I_X = 3k_4 I_D$$

Thus,

$$3k_4 I_D = -I_D$$

$$3k_4 = -1$$

$$k_4 = -\frac{1}{3}$$

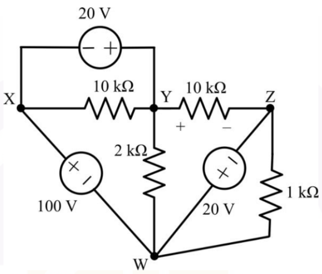
Hence,

$$k_1 = 1, \qquad k_4 = -\frac{1}{3}$$

with no restrictions on k_2 and k_3.

$$\boxed{\text{Option (A)}}$$

Solution for Q24

Q.24	For the circuit shown, which one of the following options correctly identifies the Thevenin's equivalent parameters between nodes Y and Z? 
(A)	$V_{TH} = 100$ V, $R_{TH} = 10$ kΩ
(B)	$V_{TH} = 140$ V, $R_{TH} = 0$ Ω
(C)	$V_{TH} = 100$ V, $R_{TH} = 0$ Ω

Given: Find the Thevenin equivalent between terminals Y and Z.

Step 1: Determine the Thevenin Voltage V_{TH}

Choose node W as the reference node:

$$V_W = 0$$

From the 100 V source,

$$V_X = 100 \text{ V}$$

The 20 V source between nodes X and Y has polarity $(-)$ at X and $(+)$ at Y. Hence,

$$V_Y - V_X = 20$$

$$V_Y = 100 + 20 = 120 \text{ V}$$

The 20 V source between nodes W and Z has polarity $(+)$ at W and $(-)$ at Z. Therefore,

$$V_W - V_Z = 20$$

$$0 - V_Z = 20$$

$$V_Z = -20 \text{ V}$$

The Thevenin voltage is

$$V_{TH} = V_Y - V_Z$$

Rahimul Islam Ma

$$V_{TH} = 120 - (-20)$$

$$\boxed{V_{TH} = 140 \text{ V}}$$

Step 2: Determine the Thevenin Resistance R_{TH}
Deactivate all independent voltage sources by replacing them with short circuits.

- 100 V source $\rightarrow$ short circuit
- 20 V source $\rightarrow$ short circuit
- 20 V source $\rightarrow$ short circuit

After shorting the voltage sources:

$$X \equiv Y \equiv W \equiv Z$$

Thus, terminals Y and Z are directly connected by an ideal short circuit.
Therefore,

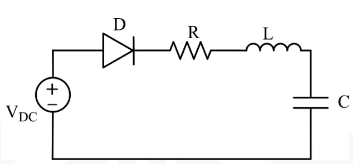
$$\boxed{R_{TH} = 0\ \Omega}$$

Hence, the Thevenin equivalent parameters are

$$\boxed{V_{TH} = 140 \text{ V}, \qquad R_{TH} = 0\ \Omega}$$

$$\boxed{\text{Option (B)}}$$

SOLUTION FOR Q23

Q.23	A circuit with ideal elements is shown.  Which one of the following options correctly identifies all the linear elements in the circuit?
(A)	R only
(B)	R, L, and C only
(C)	D only
(D)	L, C, and D only

Given: A circuit consisting of an ideal DC voltage source V_{DC}, diode D, resistor R, inductor L, and capacitor C.

A circuit element is said to be **linear** if it satisfies the principles of superposition and homogeneity.

- **Resistor** (R):
$$v = Ri$$
which is a linear relationship. Hence, R is linear.
- **Inductor** (L):
$$v = L\frac{di}{dt}$$
which is linear. Hence, L is linear.
- **Capacitor** (C):
$$i = C\frac{dv}{dt}$$
which is linear. Hence, C is linear.
- **Diode** (D): The current-voltage characteristic of an ideal diode is nonlinear:
$$i = \begin{cases} 0, & v < 0, \\ \infty, & v \geq 0. \end{cases}$$
Therefore, the diode is a nonlinear element.

Hence, the linear elements in the circuit are

$$\boxed{R,\ L,\ \text{and } C}$$

Therefore,

$$\boxed{\text{Option (B): } R,\ L, \text{ and } C \text{ only}}$$

SOLUTION FOR Q14

Q.14	Consider the following differential equation: $t^2\frac{d^2y}{dt^2} + 7t\frac{dy}{dt} + 8ty = 10\sin(t)$ Which one of the following options is correct?
(A)	It is a linear differential equation
(B)	It is a nonlinear differential equation
(C)	It is a time-invariant differential equation
(D)	It is a second-order partial differential equation

Given:

$$t^2\frac{d^2y}{dt^2} + 7t\frac{dy}{dt} + 8ty = 10\sin(t)$$

Rahimul Islam Mazumdar

To determine the nature of the differential equation, we examine the dependent variable y and its derivatives.

A differential equation is **linear** if:

- The dependent variable y and its derivatives appear only to the first power.
- There are no products of y and its derivatives.
- The coefficients may be functions of the independent variable t.

In the given equation,

$$t^2\frac{d^2y}{dt^2} + 7t\frac{dy}{dt} + 8ty = 10\sin(t),$$

the terms y, $\frac{dy}{dt}$, and $\frac{d^2y}{dt^2}$ appear linearly, and the coefficients t^2, $7t$, and $8t$ are functions of t only.

Therefore, the equation is a **linear ordinary differential equation**.

Checking the remaining options:

(B) Nonlinear differential equation **False**

(C) Time-invariant differential equation **False**
The coefficients explicitly depend on t, hence the system is time-varying.

(D) Second-order partial differential equation **False**
The equation contains ordinary derivatives, not partial derivatives.

$$\boxed{\text{The given equation is a linear differential equation.}}$$

$$\boxed{\text{Option (A)}}$$

SOLUTION FOR Q13

Given:

$$Y(s) = \frac{100}{s(s+100)}$$

Using partial fraction expansion,

$$\frac{100}{s(s+100)} = \frac{A}{s} + \frac{B}{s+100}$$

Multiplying both sides by $s(s+100)$,

$$100 = A(s+100) + Bs$$

Setting $s = 0$,

$$100 = 100A$$

$$A = 1$$

Q.13	The Laplace transform of the step response of a system is given by $$Y(s) = \frac{100}{s(s+100)}$$ The rise time is defined as the time taken for the response to go from 0.1 to 0.9 of its final value. The settling time is defined as the time taken for the response to reach 0.98 of its final value. For this system, the rise time (T_r), settling time (T_s), and time constant (T_c), all expressed in seconds, are
(A)	$T_r = 0.022$, $T_s = 0.04$, $T_c = 0.01$
(B)	$T_r = 0.22$, $T_s = 0.404$, $T_c = 0.01$
(C)	$T_r = 2.2$, $T_s = 4.04$, $T_c = 1.01$
(D)	$T_r = 22$, $T_s = 40.4$, $T_c = 10.1$

Setting $s = -100$,

$$100 = -100B$$

$$B = -1$$

Therefore,

$$Y(s) = \frac{1}{s} - \frac{1}{s+100}$$

Taking inverse Laplace transform,

$$y(t) = 1 - e^{-100t}$$

Comparing with the standard first-order response

$$y(t) = 1 - e^{-t/\tau},$$

the time constant is

$$\tau = \frac{1}{100} = 0.01 \text{ s}.$$

Hence,

$$\boxed{T_c = 0.01 \text{ s}}$$

Rise Time (T_r):

The rise time is defined as the time required for the response to rise from 10% to 90% of its final value.

At 10%,

$$0.1 = 1 - e^{-100t_{10}}$$

$$e^{-100t_{10}} = 0.9$$

$$t_{10} = -\frac{\ln(0.9)}{100} = 0.00105 \text{ s}$$

At 90%,

$$0.9 = 1 - e^{-100t_{90}}$$

$$e^{-100t_{90}} = 0.1$$

$$t_{90} = -\frac{\ln(0.1)}{100} = 0.02303 \text{ s}$$

Therefore,

$$T_r = t_{90} - t_{10}$$

$$T_r = 0.02303 - 0.00105$$

$$T_r = 0.02198 \approx 0.022 \text{ s}$$

$$\boxed{T_r = 0.022 \text{ s}}$$

Settling Time (T_s):

The settling time is defined as the time required to reach 98% of the final value.

$$0.98 = 1 - e^{-100T_s}$$

$$e^{-100T_s} = 0.02$$

$$T_s = -\frac{\ln(0.02)}{100}$$

$$T_s = \frac{3.912}{100}$$

$$T_s = 0.03912 \approx 0.04 \text{ s}$$

$$\boxed{T_s = 0.04 \text{ s}}$$

Thus,

$$\boxed{T_r = 0.022 \text{ s}, \quad T_s = 0.04 \text{ s}, \quad T_c = 0.01 \text{ s}}$$

$$\boxed{\text{Option (A)}}$$

Rahimul Islam Mazumdar